Zomato Data analysis using Python

step 1-import libraries

In [50]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

step 2- Create the data frame

In [51]:
```
dataframe = pd.read_csv('Zomato data .csv')
```

In [52]:
```
dataframe
```

Out[52]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(typ
0	Jalsa	Yes	Yes	4.1/5	775	800	Buff
1	Spice Elephant	Yes	No	4.1/5	787	800	Buff
2	San Churro Cafe	Yes	No	3.8/5	918	800	Buff
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	Buff
4	Grand Village	No	No	3.8/5	166	600	Buff
...	...	...	...	...	...	...	
143	Melting Melodies	No	No	3.3/5	0	100	Dinir
144	New Indraprasta	No	No	3.3/5	0	150	Dinir
145	Anna Kuteera	Yes	No	4.0/5	771	450	Dinir
146	Darbar	No	No	3.0/5	98	800	Dinir
147	Vijayalakshmi	Yes	No	3.9/5	47	200	Dinir

148 rows × 7 columns

In [ ]:
```
```

convert the data type of column and remove the denominator- rate

In [53]:
```
def handleRate(value):
    value = str(value).split('/')
    value = value[0];
    return float(value)


dataframe['rate'] = dataframe['rate'].apply(handleRate)
print(dataframe.head())
```

```
                  name online_order book_table  rate  votes  \
0                Jalsa          Yes        Yes   4.1    775
1        Spice Elephant          Yes         No   4.1    787
2       San Churro Cafe          Yes         No   3.8    918
3  Addhuri Udupi Bhojana           No         No   3.7     88
4          Grand Village           No         No   3.8    166

   approx_cost(for two people) listed_in(type)
0                           800          Buffet
1                           800          Buffet
2                           800          Buffet
3                           300          Buffet
4                           600          Buffet
```

Summary of the dataframe

In [54]:
```
dataframe.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   name                         148 non-null    str
 1   online_order                 148 non-null    str
 2   book_table                   148 non-null    str
 3   rate                         148 non-null    float64
 4   votes                        148 non-null    int64
 5   approx_cost(for two people)  148 non-null    int64
 6   listed_in(type)              148 non-null    str
dtypes: float64(1), int64(2), str(4)
memory usage: 8.2 KB
```

conclusion: There is no null value in dataframe Type of restaurant

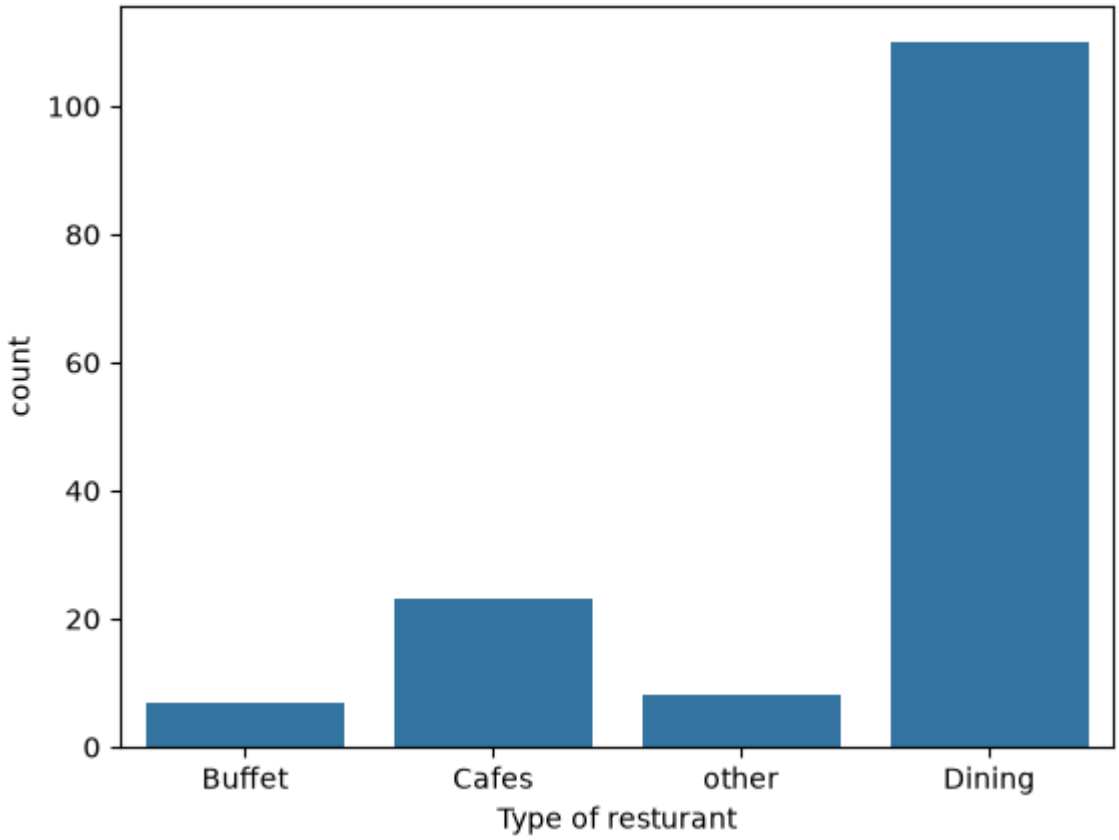
In [55]:
```
dataframe.head()
```

Out[55]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

In [56]:
```
sns.countplot(x = dataframe['listed_in(type)'])
plt.xlabel('Type of resturant')
```

Out[56]:
```
Text(0.5, 0, 'Type of resturant')
```



conclusion - majority of the resturant falls in dinning category Dining Restarurant are preferred by a larger number of individuals

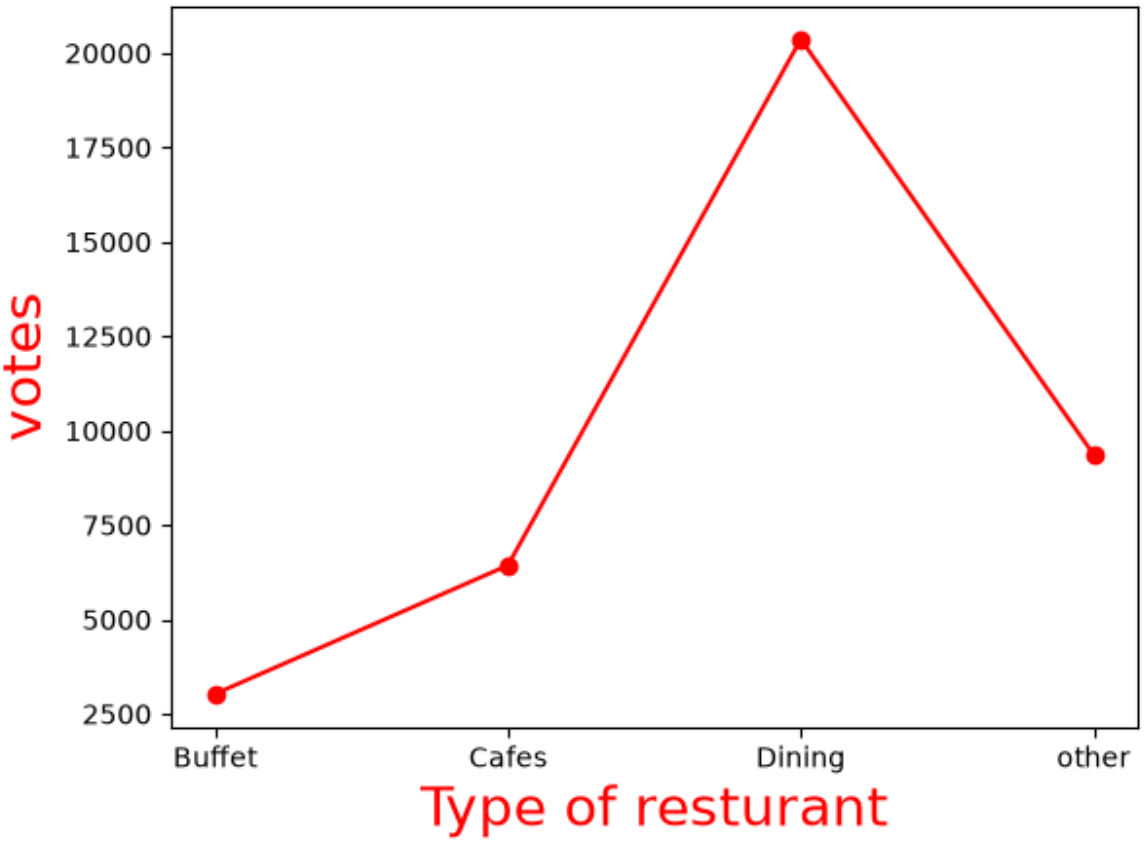
In [57]:
```
dataframe.head()
```

Out[57]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

In [58]:
```
grouped_data = dataframe.groupby('listed_in(type)')['votes'].sum()
result = pd.DataFrame({'votes':grouped_data})
plt.plot(result, c = 'red',marker = 'o')
plt.xlabel('Type of resturant', c = 'red', size = 20)
plt.ylabel('votes', c = 'red', size = 20)
```

Out[58]:
```
Text(0, 0.5, 'votes')
```



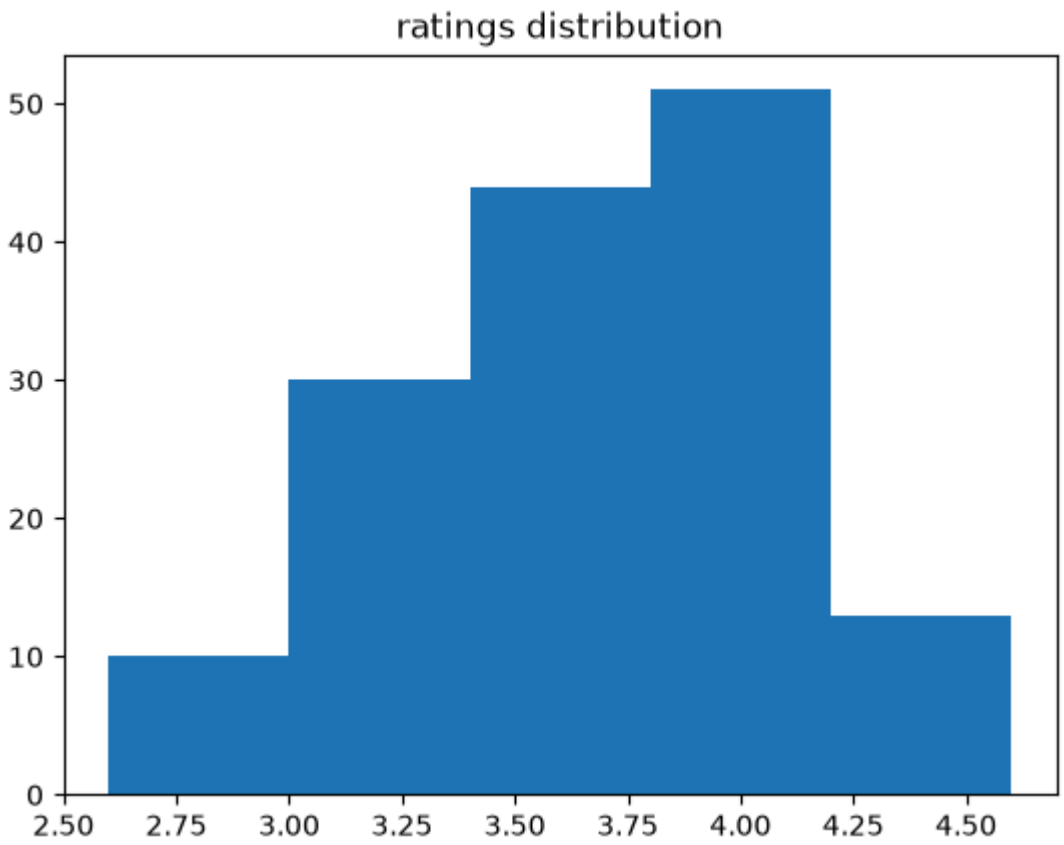
conclusion - dinning resturant has recieved maximum votes

In [59]:
```
dataframe.head()
```

Out[59]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

In [60]:
```
plt.hist(dataframe['rate'],bins = 5)
plt.title('ratings distribution')
plt.show()
```



conclusion - the majority resturant recived rating from 3.5 to 4

The majority of couples prefer restarurants with an approximate cost of 300 rupees

In [ ]:

average order spending by couples

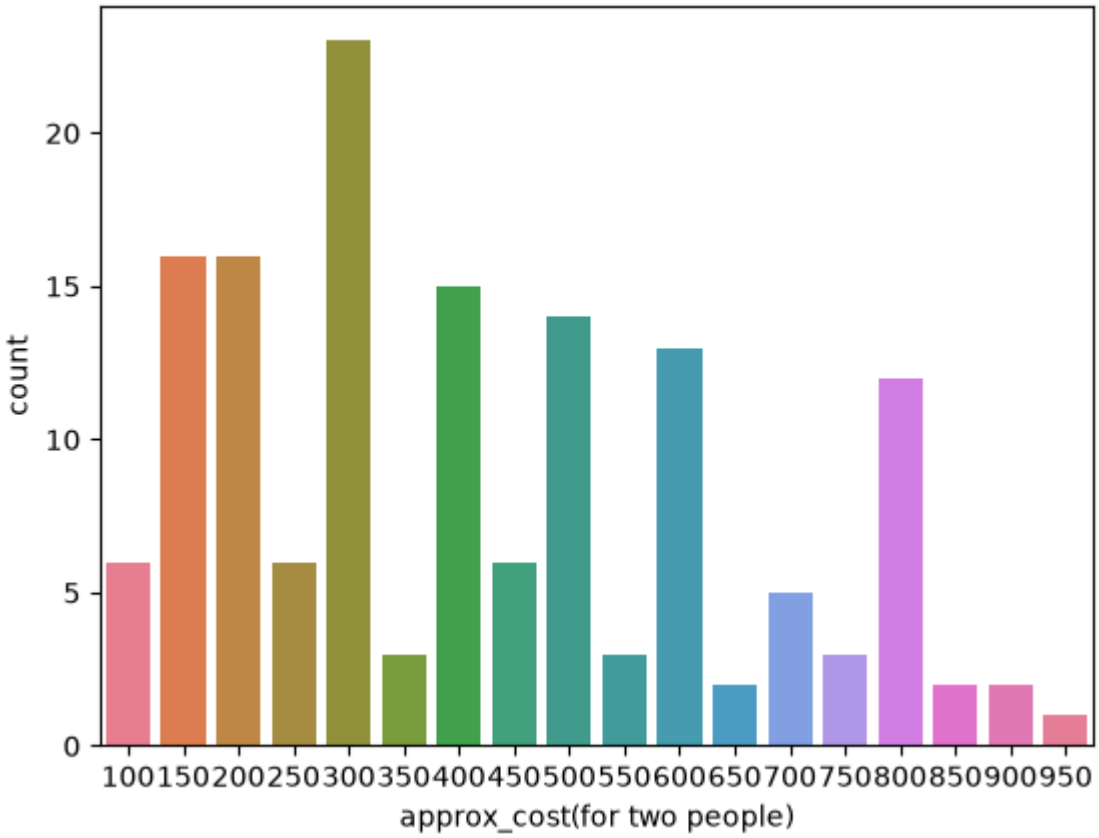
In [61]:
```
dataframe.head()
```

Out[61]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

In [64]:
```
couple_date = dataframe['approx_cost(for two people)']
sns.countplot(x = couple_date,hue = couple_date, palette = 'husl', legend = Fals
```

Out[64]:
```
<Axes: xlabel='approx_cost(for two people)', ylabel='count'>
```



conclusion - the majority of couples prefrr resturants with an approximate cost of 300 rupees

In [ ]:

which mode receives maximum rating

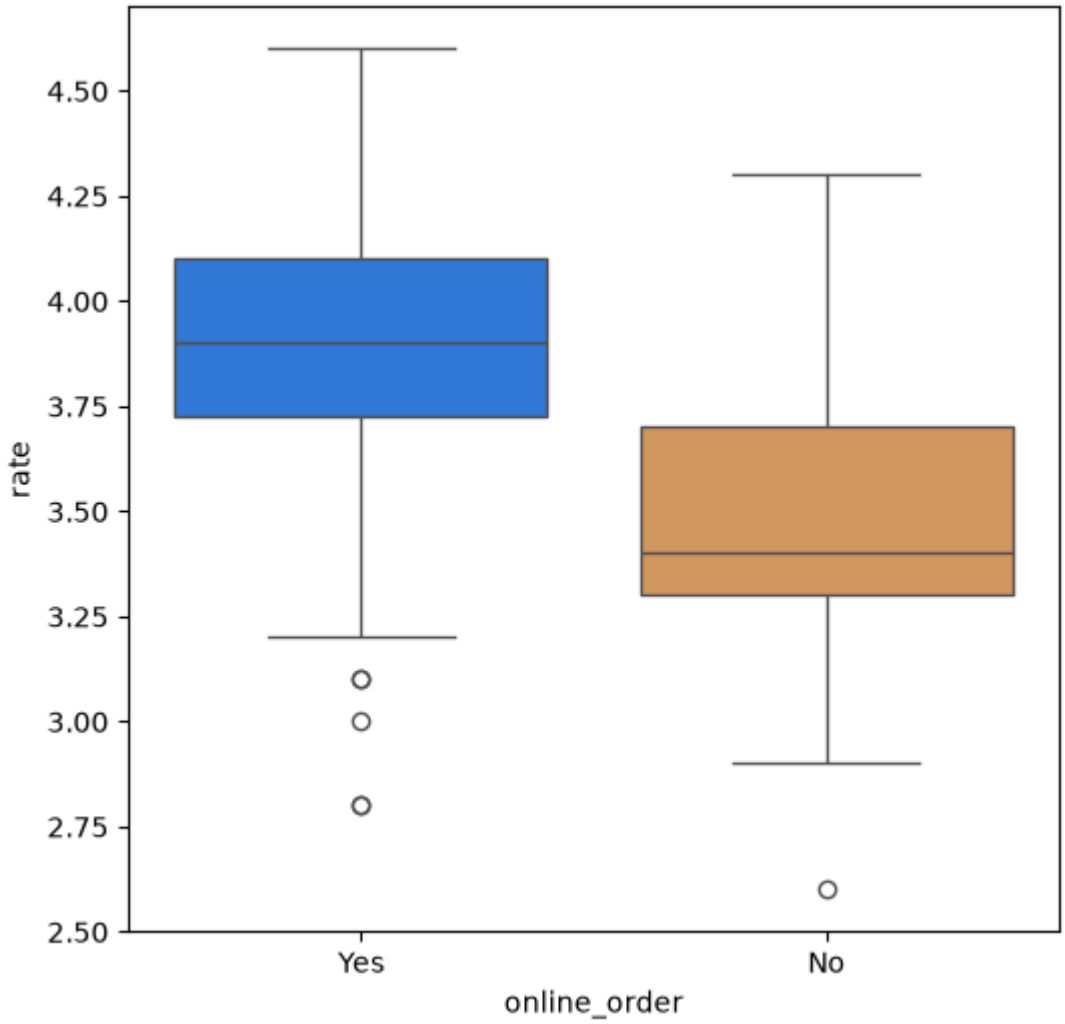
In [32]:
```
dataframe.head()
```

Out[32]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

In [68]:
```
plt.figure(figsize = (6,6))
sns.boxplot(x = 'online_order', y = 'rate', data = dataframe, hue = 'online_orde
```

Out[68]: <Axes: xlabel='online_order', ylabel='rate'>



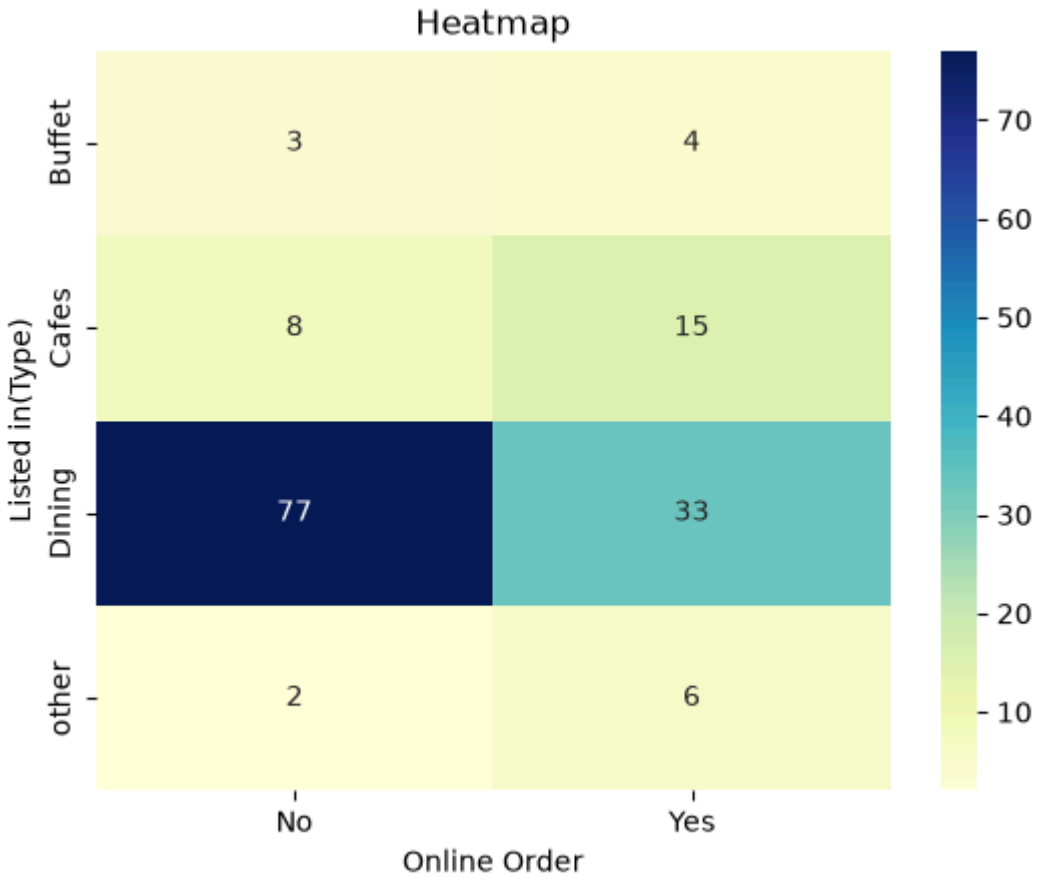
conclusion - offline order received lower rating in comparision to online order

In [34]:
```
dataframe.head()
```

Out[34]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

In [39]:
```
pivot_table = dataframe.pivot_table(index = 'listed_in(type)',columns = 'online_
sns.heatmap(pivot_table, annot = True,  cmap = 'YlGnBu',fmt = 'd')
plt.title('Heatmap')
plt.xlabel('Online Order')
plt.ylabel('Listed in(Type)')
plt.show()
```



conclusion : Dining restarurants primarily accept offline orders,whereas cafes primarily receive online orders.This suggests that clients prefer orders in person at restaurants,but prefer online ordering at cafes

In [ ]: